CNNP- Skill 2

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Section 11

2520090024

THREE-POINT CAPTURE – Analyze packet captures from three points in a network path to understand packet flow and identify performance differences.

```
raghuram32@raghuramreddy:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
70 packages can be upgraded. Run 'apt list --upgradable' to see them.
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
tshark is already the newest version (4.6.4-1).
iperf3 is already the newest version (3.20-2).
mtr-tiny is already the newest version (0.95-1.1ubuntu2).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
wireshark is already the newest version (4.6.4-1).
netcat-openbsd is already the newest version (1.234-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 70
```

```
raghuram32@raghuramreddy:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ip link show
1: lo: <LOOPBACK> mtu 65536 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
8: veth-h1@if7: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 9e:d7:51:23:da:46 brd ff:ff:ff:ff:ff:ff link-netns router
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
raghuram32@raghuramreddy:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
raghuram32@raghuramreddy:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
raghuram32@raghuramreddy:~$ h
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
h: command not found
raghuram32@raghuramreddy:~$
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
RTNETLINK answers: File exists
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
RTNETLINK answers: File exists
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
RTNETLINK answers: File exists
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
RTNETLINK answers: File exists
raghuram32@raghuramreddy:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.716 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.073 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.071 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2040ms
rtt min/avg/max/mdev = 0.071/0.286/0.716/0.303 ms
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.716 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.073 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.071 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2040ms
rtt min/avg/max/mdev = 0.071/0.286/0.716/0.303 ms
raghuram32@raghuramreddy:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
RTNETLINK answers: File exists
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
RTNETLINK answers: File exists
raghuram32@raghuramreddy:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.716 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.073 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.071 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2040ms
rtt min/avg/max/mdev = 0.071/0.286/0.716/0.303 ms
raghuram32@raghuramreddy:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1
raghuram32@raghuramreddy:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 4799271325190793666
raghuram32@raghuramreddy:~$
```

```
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
RTNETLINK answers: File exists
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
RTNETLINK answers: File exists
raghuram32@raghuramreddy:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.716 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.073 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.071 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2040ms
rtt min/avg/max/mdev = 0.071/0.286/0.716/0.303 ms
raghuram32@raghuramreddy:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1
raghuram32@raghuramreddy:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 4799271325190793666
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
6
```

```
raghuram32@raghuramreddy:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ip route
sudo ip netns exec host2 ip route
10.0.1.0/30 dev veth-h1 proto kernel scope link src 10.0.1.1
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
```[cite: 6]
Error: Nexthop has invalid gateway.
> ^C
raghuram32@raghuramreddy:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
RTNETLINK answers: File exists
Error: Nexthop has invalid gateway.
```
```